

1. Design and implement an 8-bit ALU on FPGA with inputs A[7:0], B[7:0], and control signal OPCODE[2:0]. The ALU shall support logical operations (NOT, AND, OR, XOR), arithmetic operations (8-bit addition and subtraction), and a 4×4 multiplication. Use an 8-to-1 multiplexer to select the final output based on the opcode. Implement the design in Verilog and verify its functionality through simulation.

